MIL-C-52299C

31 August 1983

SUPERSEDING
MIL-C-52299B
10 March 1970

MILITARY SPECIFICATION

CLAMP ASSEMBLY, TRESTLE BRACING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a clamp assembly and its components to be used with a trestle assembly (MIL-T-52150).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13207E9719 - Clamp Assembly, Trestle Bridge.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The clamp assembly and components shall be as shown on Top Assembly TA13207E9719 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the clamp assembly and components. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Material shall be new and unused.

3.3.1 Material deterioration and control. The clamp assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration

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that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Identification marking. The clamp assembly shall be metal die stamped as shown on the applicable drawing. Die stamping shall be legible after printing. When the identification marking is cast, the marking may be located other than as shown, provided it is on the outside surface of the movable half of the column clamp.

3.5 Treatment and painting. The clamp assembly and components, except pins and threaded portions of components, shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. The contact surfaces of all dissimilar metals shall be painted prior to assembly.

3.6 Workmanship. All holes shall be drilled to full size or drilled undersize and reamed to full size through steel templates. All burrs and projections shall be removed from holes and edges of components. Components shall be free from porosity, fins, twists, bends, and cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Components and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article clamp assembly or components, as applicable, shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The clamp assembly or components, as applicable, shall be examined for the following defects:

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.

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103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Components missing or not as specified.
106. Workmanship not as specified.
107. Castings containing blowholes, porosity, hard spots, shrinkage defects, cracks, or not smooth and well cleaned. (Minor imperfections not affecting strength may be corrected by welding).
108. Any part or component not in accordance with the QAP data requirements as shown on the drawings.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.
203. Dimensions other than those identified by QAP data not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109. Materials and containers not as specified for level A or B.
110. Unpainted and threaded surfaces not coated for level A.
111. Blocking, bracing, or securing not provided to prevent movement for level A or B.
112. Strapping not as specified for level A.
113. Gross weight or size exceeds the limitation of the fiberboard box for level B.
114. Fiberboard box not sealed and reinforced as specified for level B.

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115. Preservation and packing not in accordance with the referenced document for commerical.
116. Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Unpainted and threaded surfaces of clamp assemblies, or clamp assembly bolts when procured separately from the clamp assembly, shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.2 Commercial. Clamp assemblies, or clamp assembly bolts when procured separately from the clamp assembly, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Ten complete clamp assemblies shall be packed in a close-fitting box conforming to PPP-B-621, class 2, style 2. Blocking and bracing to prevent movement and damage shall be provided and shall be in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2) strapping shall be finish A.

5.2.1.1 Clamp assembly bolts. When procured separately from the clamp assembly, the bolts shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. The bolts shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be as specified in 5.2.1.

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5.2.2 Level B. Ten complete clamp assemblies shall be packed as specified for level A, except the box shall be class 1 and strapping shall be finish A.

5.2.2.1 Clamp assembly bolts. Clamp assembly bolts procured separately from the clamp assembly shall be packed together in close-fitting boxes conforming to PPP-B-621, class 1, style optional; PPP-B-601, domestic type, style optional; or PPP-B-636, V3c, style optional. The bolts shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The gross weight or size of fiberboard boxes shall not exceed the limitations of the box specification, and the boxes shall not exceed the limitations of the box specification, and the boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification. Wood boxes shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Clamp assemblies or clamp assembly bolts, when procured separately, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The clamp assembly and its components are intended for use in floating and fixed bridges for connecting the struts to the columns of the trestle assembly.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval and the number of units required (see 3.2).
c. Degree of preservation and degree of packing required (see 5.1 and 5.2).
d. When other than finish B strapping is required (see 5.2.1 and 5.2.1.1).

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6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced clamp assembly. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the document's first article.

6.4 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of dificiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definition.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52299C	Clamp Assembly, Trestle Bracing

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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